

HPC-Accelerated Quantum Kernel Simulation for Biomedical Data Analysis

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Abstract

Quantum-kernel methods provide a mechanism for embedding classical data into quantum state spaces, but classical simulation of quantum circuits becomes increasingly expensive as circuit size and workload scale. This work develops and evaluates a high-performance computing workflow for quantum-kernel simulation using a biomedical electromyography workload derived from the NinaPro DB2 dataset. Surface-EMG recordings were processed into 36 classical features and reduced to eight quantum input features for an eight-qubit, manually constructed ZZFeatureMap. A quantum kernel was then computed from pairwise state overlaps.

The implementation was validated across local CPU and CUDA-capable GPU simulation backends, with a maximum numerical difference of approximately 4.77×10^{-7} between the resulting kernel matrices. For the complete 102×102 eight-qubit NinaPro kernel workload, the local CPU completed the benchmark faster than the GPU, while separate qubit-scaling experiments showed that the GPU was faster at the largest tested problem size of 16 qubits. A depolarizing-noise experiment across five noise strengths showed a gradual reduction in mean state fidelity while per-sample simulation runtime remained approximately constant. Final Bridges-2 strong-scaling experiments reduced mean runtime from 0.168016 s with one MPI process to 0.024862 s with eight processes, corresponding to a speedup of approximately $6.76\times$ and a parallel efficiency of 84.47%. A final full-capacity real-data experiment used 1,404 distinct non-overlapping NinaPro EMG windows, producing 986,310 quantum-kernel pairs. Mean runtime decreased from 4.198034 s with one MPI process to 0.378437 s with 128 processes on one node. When the same 128 processes were distributed across 16

physical Bridges-2 nodes, mean runtime decreased further to 0.365481 s, corresponding to an overall **11.49** $\times$ speedup relative to the single-process baseline. On Bridges-2, V100 statevector simulation remained approximately at CPU performance through 14 qubits but achieved a measured **1.067** $\times$ speedup at the largest tested size of 16 qubits, while the GPU tensor-network backend did not outperform CPU statevector simulation over the tested range. Together, these experiments establish a reproducible framework for studying CPU, GPU, noise, and distributed-memory scaling of simulated quantum kernel workloads on biomedical data.

Keywords: quantum computing, quantum kernels, high-performance computing, MPI, GPU acceleration, biomedical signal processing, electromyography, NinaPro

1 Introduction

Quantum computing provides new computational models for representing and processing information, including approaches in which classical feature vectors are encoded into quantum states. Quantum kernel methods use these state representations to construct similarity measures in a quantum feature space, allowing classical machine-learning procedures to operate on kernel matrices derived from quantum circuits. Quantum feature maps such as the ZZFeatureMap have therefore become an important component of quantum machine-learning research [1].

For near-term research, however, quantum algorithms are frequently investigated through classical simulation rather than exclusively on physical quantum hardware. Frameworks such as Qiskit provide programmable circuit construction, while Qiskit Aer provides multiple classical simulation methods and configurable noise models [2, 3]. Classical quantum simulation is computationally demanding because the state space represented by a general statevector grows exponentially with the number of qubits. This makes simulator performance, hardware acceleration, and parallel execution important concerns when quantum experiments are extended to larger circuits or larger collections of independent simulations.

High-performance computing provides several mechanisms for addressing these computational demands. CPU parallelism can accelerate independent calculations, GPU backends can exploit highly parallel numerical hardware, and distributed-memory programming with the Message Passing Interface (MPI) can divide larger computational workloads across multiple processes. These approaches do not guarantee improved performance for every problem size, because initialization, communication, data movement, and accelerator overhead can outweigh the benefits of parallel execution for smaller workloads. Performance must therefore be measured experimentally across problem sizes and computing environments.

This work investigates these questions through a reproducible quantum-kernel benchmarking workflow. The study focuses on four related objectives: (1) validating a manually constructed quantum feature map and quantum-kernel implementation; (2) comparing CPU and CUDA-capable GPU simulation performance; (3) evaluating the

sensitivity of simulated quantum states to controlled depolarizing noise; and (4) characterizing distributed-memory scaling using MPI on the Bridges-2 high-performance computing system.

To provide a real biomedical workload, the final benchmark uses surface electromyography (sEMG) data from the NinaPro DB2 dataset. The selected data are processed into time-domain EMG features and reduced to eight numerical inputs, which are encoded by an eight-qubit quantum feature map. The resulting 102×102 quantum kernel provides a fixed real-data workload for numerical validation and CPU/GPU benchmarking, while the independent pairwise kernel calculations also provide a natural unit of work for MPI distribution.

The project originated as one component of a broader DREU-QIS research effort investigating quantum-enhanced and privacy-aware machine-learning workflows. Within that effort, the present work addresses the HPC benchmarking and parallelization track. Earlier development established the circuit-simulation framework, a SLURM execution workflow on Bridges-2, and preliminary MPI scaling. The present study extends that foundation with the NinaPro real-data workload, local CPU/GPU validation, qubit-scaling experiments, and controlled depolarizing-noise analysis. Final strong- and weak-scaling experiments using the 102-sample NinaPro quantum-kernel workload were subsequently executed on Bridges-2 using 1, 2, 4, and 8 MPI processes. The distributed-memory evaluation was then extended to matched single-node and multi-node configurations at 8, 16, and 32 MPI processes. A larger balanced real-data workload using seven non-overlapping windows from each gesture-repetition segment produced 714 distinct real EMG windows and 255,255 kernel pairs.

To evaluate the MPI implementation at the maximum non-overlapping-window capacity of the selected recording, a final workload retained every complete 512-sample window available from each gesture-repetition segment. This produced 1,404 distinct real EMG windows and 986,310 upper-triangular quantum-kernel pairs. Process-count scaling was evaluated through 128 MPI ranks, and physical-node scaling was evaluated by distributing 128 fixed ranks across as many as 16 Bridges-2 Regular Memory nodes.

The primary contribution of this work is therefore not a claim of quantum advantage or improved biomedical classification accuracy. Instead, it is an experimental characterization of how a reproducible simulated quantum-kernel workload behaves across CPU, GPU, noisy-simulation, and distributed HPC configurations. This distinction is important because simulator performance and parallel scalability are prerequisites for evaluating larger quantum machine-learning experiments reliably.

2 Related Work

2.1 Quantum Feature Maps and Quantum Kernels

Quantum kernel methods encode classical data into quantum states and use state-space similarity as a kernel for downstream machine-learning methods. Havlíček et al. demonstrated a quantum-enhanced feature-space approach in which classical samples are mapped into a Hilbert space through parameterized quantum circuits and compared using quantum-state overlaps [1]. This framework motivates the use of quantum feature maps as a mechanism for constructing kernel matrices from classical data.

In the present work, this general approach is used as a computational workload rather than as evidence of quantum classification advantage. The focus is on the cost of constructing and simulating the feature-map states and evaluating their pairwise overlaps across different classical computing environments.

2.2 Quantum-Circuit Simulation

Practical quantum-algorithm research frequently relies on classical simulation during development and validation. Qiskit provides a software framework for constructing, transforming, and evaluating quantum circuits [2]. Qiskit Aer extends this workflow with high-performance simulation methods including statevector and density-matrix simulation, noise models, and accelerator-capable execution [3].

Classical simulation becomes increasingly expensive as the number of qubits increases because the size of a general statevector grows exponentially with qubit count. Simulator implementation, available memory, CPU parallelism, and accelerator hardware therefore become important experimental considerations. These characteristics motivate the CPU/GPU and qubit-scaling measurements used in this study.

2.3 High-Performance Computing and MPI

The Message Passing Interface (MPI) is a standardized programming model for distributed-memory parallel computing that enables independent processes to coordinate computation and exchange data across processors and compute nodes [4, 5].

Parallel performance is commonly characterized through strong and weak scaling. Strong scaling measures how runtime changes as additional processing resources are applied to a fixed total workload. Weak scaling instead increases the total problem size with the amount of available parallel hardware in order to evaluate whether runtime remains stable as the system scales.

The present work applies MPI at the quantum-kernel workload level. Independent pairwise kernel calculations are distributed between ranks rather than partitioning the internal quantum state of one circuit simulation. This design is well suited to a kernel matrix because different matrix elements can be computed independently before the results are combined.

2.4 Distributed and Accelerated Quantum Simulation

Quantum simulation frameworks can also use parallel hardware internally. Qiskit Aer supports configurable parallel execution and provides mechanisms for GPU-accelerated and MPI-enabled simulation in supported environments [6]. Such distributed simulator implementations can partition the simulation of a single quantum state when the circuit becomes too large or computationally expensive for one process.

The strategy used in this project differs from that model. Local GPU execution accelerates individual state simulations, while MPI distributes independent quantum-kernel pair calculations. This provides two distinct levels of parallelism: acceleration of the simulation kernel itself and distribution of the larger collection of simulations that form the quantum-kernel matrix.

Recent work by Brown et al. examines distributed multi-GPU quantum-circuit simulation on HPC systems and introduces MPI support into the QED-C application-oriented benchmark suite [7]. Their results emphasize that inter-GPU communication and network performance can become major bottlenecks when state-vector simulation is distributed across multiple GPUs and nodes. This demonstrates that quantum-simulation performance depends not only on accelerator throughput, but also on the communication architecture connecting the available computing resources.

The MPI strategy used in the present work differs from the simulator-level parallelism evaluated by Brown et al. Rather than distributing a single statevector across multiple GPUs, the QuantumHPC implementation distributes independent quantum-kernel pair calculations across MPI ranks. This provides a task-level parallelism strategy in which the individual quantum simulations remain local to each process.

2.5 Biomedical EMG Data and NinaPro

Surface electromyography records electrical activity associated with muscle activation and is commonly used for studying hand and gesture recognition. The NinaPro project provides benchmark datasets containing synchronized surface-EMG and movement information for research involving natural hand movements and prosthetic-control systems [8].

NinaPro DB2 includes recordings acquired using multiple surface-EMG channels across repeated hand movements. In this project, the dataset is not used to claim improved gesture-recognition performance. Instead, it provides a repeatable real-data source from which numerical feature vectors can be constructed for quantum-kernel benchmarking.

Using recorded biomedical data rather than exclusively synthetic vectors also allows the HPC workflow to be evaluated with a realistic preprocessing stage that includes segmentation, time-domain EMG feature extraction, dimensionality reduction, and quantum feature encoding.

2.6 Relationship to the Present Study

Previous work establishes the foundations for quantum feature spaces, quantum simulation, accelerator-supported execution, distributed-memory computing, and biomedical EMG datasets. The present study connects these components into a single reproducible performance experiment.

The principal focus is the computational behavior of a quantum-kernel workload derived from biomedical data. Rather than evaluating only one simulator or one hardware configuration, the study compares CPU and GPU execution, examines run-time growth with qubit count, evaluates state fidelity under controlled depolarizing noise, and applies MPI strong- and weak-scaling experiments to the pairwise kernel computation.

This combination allows the study to distinguish three related questions: whether independently implemented simulation backends produce the same quantum kernel, when GPU acceleration becomes beneficial for the tested workload, and how effectively independent quantum-kernel calculations can be distributed across HPC resources.

3 Methodology

3.1 Experimental Framework

The QuantumHPC framework was developed as a modular workflow for constructing, simulating, and benchmarking quantum-kernel workloads across heterogeneous computing environments. The workflow separates biomedical data processing, quantum feature-map construction, quantum-state simulation, kernel evaluation, noise characterization, MPI workload distribution, and result analysis. This separation allows the same quantum workload to be evaluated under different simulation backends and computing configurations without changing the underlying data representation or kernel definition.

Qiskit is used for quantum-circuit construction and CPU-based reference simulation [2, 3]. Local GPU simulation is performed using a CUDA-capable qsim backend through a translation layer that maps the supported gates of the manually constructed Qiskit feature map to the corresponding GPU simulation representation. MPI is used as a workload-level parallelization layer for distributing independent quantum-kernel calculations across processes.

The study consists of four primary experimental components: numerical validation of CPU and GPU quantum-kernel implementations, local CPU/GPU performance and qubit-scaling benchmarks, controlled depolarizing-noise characterization, and distributed-memory strong-, weak-, and multi-node scaling experiments on Bridges-2.

3.2 NinaPro Biomedical Workload

The final application workload is derived from the NinaPro DB2 surface electromyography (sEMG) dataset [8]. The benchmark uses the Subject 1, Exercise 1, Acquisition 1 recording. The selected exercise contains 17 gesture labels, and six repetitions are used for each gesture, producing 102 gesture-repetition samples.

The recording contains 12 sEMG channels sampled at 2 kHz. For each gesture-repetition combination, the longest contiguous segment matching the corresponding gesture and repetition labels is identified. A deterministic 512-sample window is then extracted from the center of that segment. At the 2 kHz sampling rate, each selected window represents 256 ms of EMG activity.

Only gesture samples are included in the final quantum-kernel workload. Rest segments are not treated as additional gesture classes. The resulting dataset therefore contains exactly $[17 \text{ gestures} \times 6 \text{ repetitions} = 102 \text{ samples}]$.

The 102-sample workload is used as the primary fixed workload for numerical validation, local CPU/GPU benchmarking, noise characterization, and the initial MPI scaling experiments. A larger real-data MPI workload was additionally constructed to evaluate multi-node behavior at a higher computational load.

For this extension, seven distinct non-overlapping 512-sample windows were selected from each gesture-repetition segment. The shortest identified segment contained 3906 samples, which was sufficient to support seven non-overlapping 512-sample windows in every case. A centered block of seven consecutive windows was used for

each segment so that every gesture–repetition combination contributed the same number of samples. The resulting workload contained $[17 \times 6 \times 7 = 714]$ distinct real EMG windows. These windows originate from the same Subject 1, Exercise 1, Acquisition 1 recording and therefore increase computational workload size rather than the number of independent subjects or gesture repetitions.

A final full-capacity MPI workload was constructed from the same recording by retaining every complete non-overlapping 512-sample window available from the longest contiguous segment for each gesture–repetition pair. Because segment lengths differ, the number of complete windows ranged from 7 to 27 per segment, with a mean of approximately 13.76. Across all 102 gesture–repetition segments, this procedure produced 1,404 distinct real EMG windows.

The full-capacity workload maximizes the number of complete non-overlapping windows available from the selected Subject 1, Exercise 1, Acquisition 1 recording. It should not be interpreted as use of the complete NinaPro DB2 dataset across all subjects or exercises.

3.3 EMG Feature Extraction and Dimensionality Reduction

Three time-domain features are extracted independently from each of the 12 EMG channels: mean absolute value (MAV), root mean square (RMS), and zero crossings (ZC). Zero crossings are counted when consecutive nonzero samples change sign. This produces $[12 \text{ channels} \times 3 \text{ features per channel} = 36 \text{ classical features}]$ for each selected EMG window.

The 36-dimensional feature vectors are first standardized using `StandardScaler`. Principal component analysis (PCA) is then applied to reduce the representation to eight dimensions. Finally, each PCA component is rescaled using a min–max transformation to the interval $[0, \pi]$.

This produces an input matrix of shape 102×8 , with one eight-dimensional quantum feature vector for each gesture–repetition sample. The use of eight features allows the biomedical workload to be encoded directly into an eight-qubit feature map.

For the larger real-data MPI experiment, the same feature-extraction and dimensionality-reduction sequence was applied to all 714 selected windows, producing a separate 714×8 quantum-feature matrix. The larger workload therefore changes the number of real EMG windows while preserving the same 36-feature classical representation and eight-dimensional quantum input space.

The same preprocessing sequence was also applied independently to the final 1,404-window full-capacity workload, producing a 1404×8 quantum-feature matrix. All three real-data workloads therefore use the same 36-feature classical representation and eight-dimensional quantum input space; only the number and selection of real EMG windows differ.

3.4 Quantum Feature Map and Kernel

Classical feature vectors are encoded into quantum states using a manually constructed `ZZFeatureMap`-inspired quantum circuit. Following the quantum-kernel formulation described by Havlíček et al. [1], similarity between two encoded samples is defined by

the squared magnitude of their quantum state overlap. The circuit is constructed from primitive Hadamard, phase, and controlled-X gates rather than relying exclusively on a high-level feature-map constructor. This gate-level implementation was developed to provide explicit control over the simulated circuit and to support consistent translation between CPU and GPU simulation backends.

For a classical feature vector $\mathbf{x}$, the feature-map circuit prepares a quantum state

$$|\psi(\mathbf{x})\rangle.$$

For two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, the quantum kernel is defined as the squared magnitude of their state overlap,

$$K(\mathbf{x}_i, \mathbf{x}_j) = |\langle\psi(\mathbf{x}_i)|\psi(\mathbf{x}_j)\rangle|^2.$$

Evaluating the full NinaPro workload produces a symmetric 102×102 kernel matrix. Because

$$K(\mathbf{x}_i, \mathbf{x}_j) = K(\mathbf{x}_j, \mathbf{x}_i),$$

only the upper-triangular portion of the matrix, including the diagonal, must be computed explicitly. For $N = 102$ samples, this corresponds to

$$\frac{N(N+1)}{2} = \frac{102(103)}{2} = 5253$$

kernel-pair evaluations.

For the 714-window real-data MPI extension, the corresponding symmetric kernel contains

$$\frac{714(715)}{2} = 255255$$

upper-triangular kernel-pair evaluations. This represents approximately 48.6 times as many pair evaluations as the primary 102-sample kernel while retaining the same eight-qubit feature-map structure.

For the full-capacity workload of 1,404 real windows, the number of upper-triangular kernel-pair evaluations increases to $\frac{1404(1405)}{2} = 986310$.

This workload contains approximately 187.8 times as many kernel-pair evaluations as the original 102-sample benchmark while preserving the same eight-qubit feature-map construction.

The resulting matrix is validated for finite values, symmetry, a unit diagonal, and kernel values within the interval $[0, 1]$.

3.5 Local CPU and GPU Simulation

The reference CPU implementation evaluates the manually constructed feature map using Qiskit-based statevector simulation. Each feature vector is encoded once, and pairwise state overlaps are then used to construct the kernel matrix.

A separate CUDA-capable GPU backend was implemented using qsim [9]. The manually constructed Qiskit circuit is translated into the corresponding supported

qsim operations before execution on the GPU. This approach allows the same feature-map structure and input data to be evaluated by independent CPU and GPU simulation implementations.

The local system used for these experiments contains an Intel Core i7-9700K CPU and an NVIDIA GeForce RTX 2080 SUPER GPU. Numerical validation is performed by constructing the complete 102×102 NinaPro kernel matrix with both backends and comparing the resulting matrices element by element. Runtime measurements are collected separately from numerical agreement so that performance differences do not affect correctness validation.

3.6 Qubit-Scaling Experiment

A separate controlled benchmark evaluates how simulator runtime changes as circuit size increases. Quantum feature maps are evaluated at 4, 6, 8, 10, 12, 14, and 16

qubits using both the CPU and GPU simulation paths.

Each configuration is executed once as a warm-up before five timed runs are recorded. Mean runtime and runtime variation are calculated for each qubit count. The purpose of this experiment is to determine whether the relative performance of CPU and GPU simulation changes as the quantum state space increases, rather than assuming that GPU acceleration is beneficial for all circuit sizes.

3.7 Depolarizing-Noise Characterization

Noise sensitivity is evaluated separately from the CPU/GPU performance benchmark. The experiment uses Qiskit Aer density-matrix simulation on the CPU and evaluates all 102 NinaPro feature vectors under depolarizing noise.

Five depolarizing-noise strengths are tested: $p \in \{0.001, 0.005, 0.010, 0.020, 0.050\}$.

For each input feature vector, the ideal pure state $|\psi\rangle$ is generated first. The same circuit is then simulated with the selected depolarizing-noise model to produce a noisy density matrix ρ . Agreement between the ideal and noisy states is measured using $F = \langle\psi|\rho|\psi\rangle$.

The experiment records the fidelity and simulation runtime for every sample and noise strength. Mean fidelity, fidelity variation, and mean runtime per sample are then calculated for each value of p .

The depolarizing-noise parameter used in this experiment represents simulated quantum hardware noise. It is distinct from the differential-privacy parameter ϵ used in earlier project experiments and should not be interpreted as a differential-privacy mechanism.

3.8 MPI Quantum-Kernel Parallelization

The quantum-kernel workload is naturally parallelizable because the state overlap for one pair of feature vectors can be evaluated independently of the overlap for another pair. The MPI implementation therefore distributes upper-triangular kernel-pair calculations across independent MPI ranks rather than using MPI to divide the internal statevector simulation of a single circuit.

This distinction separates the present approach from an MPI-enabled simulator in which one quantum state is distributed internally across multiple processes. Here, MPI operates at the workload level: each rank evaluates a subset of the kernel pairs, and the partial results are subsequently combined to reconstruct the complete symmetric kernel matrix.

The original strong- and weak-scaling experiments use $P \in \{1, 2, 4, 8\}$ MPI processes. Subsequent multi-node experiments extend the process count to 32 ranks, while the final full-capacity experiment uses as many as 128 MPI processes distributed across as many as 16 physical Bridges-2 nodes. MPI execution and communication are implemented through `mpi4py` [10], while process placement and resource allocation are managed by SLURM on Bridges-2 [11].

3.9 Strong-Scaling Experiment

Strong scaling evaluates whether a fixed problem can be completed more quickly as additional MPI processes are introduced. The complete 102-sample NinaPro workload is therefore held constant for all process counts.

The workload contains 5253 upper-triangular kernel-pair calculations. These pairs are distributed across 1, 2, 4, and 8 MPI processes. As process count increases, the approximate amount of work assigned to each process decreases while the total computational problem remains unchanged.

For each process count, wall-clock runtime is measured and compared with the single-process baseline. Strong-scaling speedup is defined as

$$S_P = \frac{T_1}{T_P},$$

where T_1 is the mean runtime using one process and T_P is the mean runtime using P processes.

Parallel efficiency is calculated as

$$E_P = \frac{S_P}{P}.$$

For presentation as a percentage, E_P is multiplied by 100.

3.10 Weak-Scaling Experiment

Weak scaling evaluates performance while the computational workload grows with the number of MPI processes. Because quantum-kernel construction requires approximately $O(N^2)$ pairwise calculations, maintaining a roughly constant number of kernel pairs per MPI process requires the sample count to grow approximately with $\sqrt{P}$.

The weak-scaling workloads are therefore defined as

MPI Processes	Workload Samples	Kernel Pairs
1	102	5253
2	144	10440
4	204	20910
8	289	41905

which keeps the number of kernel-pair calculations per process near 5200 across all four configurations.

Because the source NinaPro subset contains only 102 distinct feature vectors, larger weak-scaling workloads are generated by deterministic cyclic reuse of those real feature vectors. These repeated vectors are used only to increase the computational workload. They do not represent additional biological subjects, gesture observations, or independent biomedical samples.

Weak-scaling efficiency is evaluated relative to the single-process baseline while accounting for the small variation in kernel-pair evaluations per rank. For process count P , the reported efficiency is calculated as

$$E_P^{\text{weak}} = \frac{T_1}{T_P} \frac{W_P}{W_1},$$

where T_P is the mean runtime and W_P is the number of kernel-pair evaluations per process.

This experiment holds the kernel-pair workload per rank approximately constant, rather than every component of the complete timed workflow. The timed region also includes quantum-state encoding, MPI state exchange, kernel evaluation, and result gathering.

3.11 Multi-Node MPI Extension

A follow-on experiment was performed to distinguish MPI process-count scaling from the effects of distributing those processes across multiple physical Bridges-2 compute nodes. MPI provides a distributed-memory programming model in which processes may execute within one node or communicate across multiple nodes [4, 5].

The first topology experiment retained the fixed 102-sample NinaPro workload used for strong scaling, containing eight quantum features per sample and 5253 upper-triangular quantum-kernel pairs. Matched process counts were used to isolate the effect of node placement. Eight MPI processes were evaluated on either one node or two nodes, sixteen processes were evaluated on either one node or two nodes, and thirty-two processes were evaluated on either one node or four nodes.

Because the 102-sample workload is relatively small for a multi-node HPC system, a second experiment was performed using a larger workload constructed entirely from distinct real NinaPro EMG windows. Seven non-overlapping 512-sample windows were selected from each of the 102 gesture-repetition segments, producing 714 real windows and 255,255 kernel-pair evaluations. The larger workload was evaluated using one MPI process as a baseline and matched single-node and multi-node configurations at 8, 16, and 32 processes.

Each configuration used one untimed warm-up followed by five timed iterations. MPI rank placement was explicitly controlled so that the requested number of processes was assigned to each node.

The benchmark divides both quantum-state encoding and upper-triangular kernel-pair evaluation among MPI ranks. After local state encoding, statevectors are exchanged among the MPI processes before each rank evaluates its assigned subset of

kernel pairs. The resulting pair values are then gathered to reconstruct and validate the complete symmetric quantum-kernel matrix.

This approach evaluates workload-level distributed-memory parallelism rather than distributed simulation of a single quantum statevector. The purpose of the multi-node experiments is to determine how the relationship between computational workload size and communication overhead affects the usefulness of additional physical compute nodes.

3.12 Full-Capacity Real-Data MPI Extension

To evaluate MPI scaling using the maximum number of complete non-overlapping windows available from the selected NinaPro recording, a final real-data workload was constructed from Subject 1, Exercise 1, Acquisition 1. For each of the 102 gesture-repetition segments, every complete non-overlapping 512-sample window from the longest contiguous segment was retained. The number of available windows varied from 7 to 27 per segment, with a mean of approximately 13.76 windows per segment.

This procedure produced 1,404 distinct real EMG windows. The same MAV, RMS, and zero-crossing feature extraction, StandardScaler normalization, PCA reduction to eight dimensions, and MinMax scaling to $[0, \pi]$ were then applied. The resulting eight-qubit workload contained $\frac{1404(1405)}{2} = 986310$ upper-triangular quantum-kernel pair evaluations.

Two MPI experiments were performed. First, process-count scaling was evaluated on one Bridges-2 Regular Memory node using 1, 8, 16, 32, 64, and 128 MPI processes. Second, physical-node scaling was evaluated while holding the total process count fixed at 128 ranks. The 128 ranks were distributed across 1, 2, 4, 8, and 16 physical nodes, corresponding to 128, 64, 32, 16, and 8 ranks per node, respectively. Each configuration used one untimed warm-up followed by five timed iterations.

3.13 Bridges-2 Execution Environment

The distributed experiments were executed on the Pittsburgh Supercomputing Center Bridges-2 system. CPU MPI experiments used the Bridges-2 Regular Memory partition and SLURM for batch scheduling. One CPU core was assigned to each MPI rank, and hidden threading within individual ranks was disabled so that the requested MPI process count determined the parallel configuration.

The initial final benchmark evaluated 1, 2, 4, and 8 MPI processes for the 102-sample strong- and weak-scaling workloads. Follow-on topology experiments extended execution through 32 MPI processes and four physical nodes.

The final full-capacity real-data experiment used 1,404 distinct non-overlapping NinaPro EMG windows and extended process-count scaling to 128 MPI ranks. Single-node process scaling used 1, 8, 16, 32, 64, and 128 ranks. Physical-node scaling then held the total process count fixed at 128 ranks while distributing those ranks across 1, 2, 4, 8, and 16 Bridges-2 Regular Memory nodes. The largest tested distributed configuration therefore used 16 physical nodes with eight MPI ranks assigned to each node.

Each timed configuration used one untimed warm-up followed by five timed iterations. Requested rank placement was verified from the SLURM/MPI job logs, and output was written to CSV files for subsequent analysis.

An additional Bridges-2 accelerator experiment was configured using an NVIDIA Tesla V100-32GB GPU and Qiskit Aer with GPU support. The benchmark evaluates Aer statevector simulation on the CPU, Aer statevector simulation on the V100 GPU, and the Aer GPU tensor-network method using the installed cuQuantum software stack. The full 102-sample NinaPro kernel and controlled 4–16-qubit scaling workloads are evaluated using one untimed warm-up followed by five timed runs per configuration.

The Bridges-2 accelerator measurements are treated as a separate hardware and simulator configuration from the local RTX 2080 SUPER qsim measurements. Consequently, the results are reported as an independent HPC accelerator comparison rather than being combined directly with the local GPU timings.

3.14 Performance Metrics and Reproducibility

The primary performance metrics used throughout the study are wall-clock runtime, speedup, and parallel efficiency. Numerical validation metrics include the maximum and mean absolute difference between CPU- and GPU-generated kernel matrices, as well as symmetry, diagonal, range, and finite-value checks.

Experimental output is stored in machine-readable CSV files, while separate analysis scripts generate the figures used in the final report. The NinaPro preprocessing configuration, quantum feature-map construction, simulator configuration, MPI workload decomposition, and analysis procedures are fixed in the project repository so that the final benchmarks can be reproduced using the same experimental parameters.

To reduce startup effects in the timing measurements, benchmark configurations are executed with an untimed warm-up before repeated timed runs. MPI ranks are synchronized before timed distributed evaluations so that differences in process startup do not contaminate the measured wall time, consistent with benchmarking practices used in recent distributed quantum-simulation studies [7].

4 Results

4.1 CPU/GPU Quantum-Kernel Validation

The complete 102×102 NinaPro quantum-kernel matrix was evaluated using both the local CPU reference implementation and the CUDA-capable qsim GPU implementation. The two backends produced numerically consistent kernel matrices.

The maximum absolute difference between corresponding matrix elements was 4.768×10^{-7} , while the mean absolute difference was approximately 5.043×10^{-9} .

Both matrices satisfied the expected validation properties: all values were finite, the matrices were symmetric, diagonal entries were equal to one within numerical precision, and all kernel values remained within the interval $[0, 1]$.

These results demonstrate that the independently implemented CPU and GPU simulation paths evaluate the manually constructed quantum feature map consistently within floating-point precision.

4.2 Local NinaPro CPU/GPU Performance

The full 102×102 eight-qubit NinaPro quantum-kernel workload was then benchmarked on the local CPU and GPU systems. The CPU implementation produced a mean runtime of 0.219500 s, while the GPU implementation produced a mean runtime of 0.346955 s.

Table 1 summarizes the final benchmark measurements.

Table 1 Local CPU and GPU runtime for the complete 102×102 NinaPro quantum-kernel workload.

Backend	Mean Runtime (s)	Std. Dev. (s)	Median (s)
CPU	0.219500	0.007410	0.220389
GPU	0.346955	0.005792	0.347514

For this specific eight-qubit workload, the ratio of CPU runtime to GPU runtime was approximately $\frac{0.219500}{0.346955} \approx 0.633$. Equivalently, the GPU required approximately 1.58 times the CPU runtime for the complete kernel benchmark.

This result does not imply that GPU simulation is generally slower than CPU simulation. Rather, it indicates that the workload at eight qubits is small enough that GPU initialization, data-transfer, or execution overhead can offset the computational benefit of accelerator hardware. The separate qubit-scaling experiment provides additional context for this behavior.

4.3 Qubit-Scaling Results

The controlled qubit-scaling benchmark evaluated both simulation backends from 4 to 16 qubits. Table 2 reports the measured mean runtime and standard deviation after one warm-up execution and five timed runs for each configuration.

At qubit counts from 4 through 14, the CPU implementation completed the tested simulation faster than the GPU implementation. The relative behavior changed at 16 qubits. At that problem size, the mean CPU runtime increased to approximately 1.068 s, while the GPU mean runtime was approximately 0.438 s.

The resulting CPU-to-GPU runtime ratio at 16 qubits was approximately $\frac{1.068073}{0.438449} \approx 2.44$ indicating that the GPU completed the tested 16-qubit workload approximately 2.44 times faster than the CPU.

The observed crossover supports the interpretation that accelerator performance depends strongly on problem size. GPU overhead dominates smaller workloads, while the increasing simulation cost at larger qubit counts can provide sufficient parallel work to make GPU execution advantageous.

Table 2 Local CPU and GPU quantum-simulation runtime as a function of qubit count.

Qubits	CPU		GPU	
	Mean (s)	Std. Dev. (s)	Mean (s)	Std. Dev. (s)
4	0.013458	0.005099	0.015172	0.001294
6	0.016633	0.002741	0.019054	0.001248
8	0.020191	0.002670	0.023994	0.000710
10	0.030254	0.016823	0.031995	0.001196
12	0.034459	0.000754	0.051124	0.001226
14	0.079986	0.008856	0.122804	0.005826
16	1.068073	0.391942	0.438449	0.027035

4.4 Depolarizing-Noise Results

The noise experiment evaluated all 102 NinaPro quantum feature vectors under five depolarizing-noise strengths. Table 3 summarizes the mean state fidelity and mean simulation runtime per sample.

Table 3 NinaPro quantum-state fidelity and runtime under depolarizing noise.

Noise Strength p	Mean Fidelity	Fidelity Std. Dev.	Mean Runtime/Sample (s)
0.001	0.99996078	0.00013509	0.043027
0.005	0.99980392	0.00060754	0.042118
0.010	0.99960784	0.00135088	0.042163
0.020	0.99921569	0.00270177	0.042112
0.050	0.99803922	0.00670544	0.041740

Mean fidelity decreased monotonically as the depolarizing-noise strength increased. At $p = 0.001$, the mean fidelity was approximately 0.99996, while at $p = 0.050$ it decreased to approximately 0.99804. Fidelity variation also increased as the noise strength increased.

In contrast, the average simulation runtime remained nearly constant across the tested noise strengths, with all mean per-sample runtimes remaining close to 0.042 s. Within this experiment, increasing the depolarizing-noise probability therefore affected state fidelity substantially more than simulation runtime.

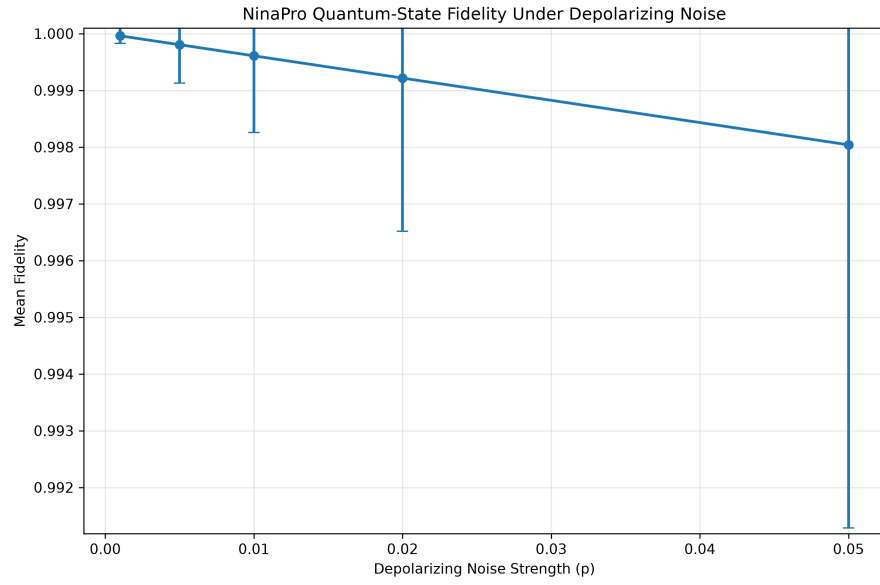


Fig. 1 Mean quantum-state fidelity for the 102-sample NinaPro workload as depolarizing-noise strength increases. Error bars represent the observed fidelity variation across samples.

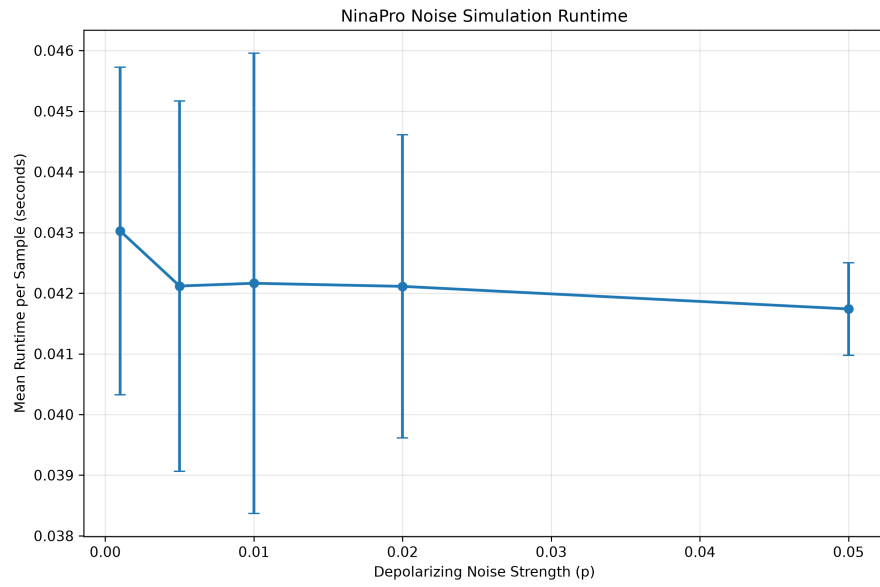


Fig. 2 Mean simulation runtime per NinaPro sample across the five tested depolarizing-noise strengths.

4.5 Preliminary Bridges-2 MPI Scaling

Before development of the final NinaPro MPI benchmark, an earlier validation workload was executed on Bridges-2 using 1, 2, and 4 MPI processes. Mean runtime decreased from approximately 6.1 s with one process to 3.5 s with two processes and 2.3 s with four processes, corresponding to preliminary speedups of approximately $1.74\times$ and $2.61\times$, respectively.

These measurements established that the SLURM and MPI execution workflow functioned correctly on Bridges-2. They are retained as preliminary validation results but are superseded by the final NinaPro scaling experiment reported below.

4.6 Final NinaPro MPI Scaling

The final distributed-memory benchmark evaluated the NinaPro quantum-kernel workflow using 1, 2, 4, and 8 MPI processes on Bridges-2. All benchmark configurations completed successfully, and every timed kernel matrix passed the numerical validation checks.

4.6.1 Strong Scaling

The strong-scaling experiment held the 102-sample, 5253-pair workload constant while increasing MPI process count. Table 4 summarizes the measured runtime, speedup, and parallel efficiency.

Table 4 Final NinaPro strong-scaling results on Bridges-2.

MPI Processes	Samples	Mean Runtime (s)	Speedup	Efficiency (%)
1	102	0.168016	1.000	100.00
2	102	0.084332	1.992	99.62
4	102	0.044457	3.779	94.48
8	102	0.024862	6.758	84.47

Mean runtime decreased from 0.168016 s with one process to 0.084332 s with two processes, 0.044457 s with four processes, and 0.024862 s with eight processes. Relative to the single-process baseline, the corresponding measured speedups were $1.992\times$, $3.779\times$, and $6.758\times$.

Parallel efficiency remained 99.62% at two processes, 94.48% at four processes, and 84.47% at eight processes. The results therefore show near-linear scaling at the lower process counts and continued substantial acceleration through eight MPI processes.

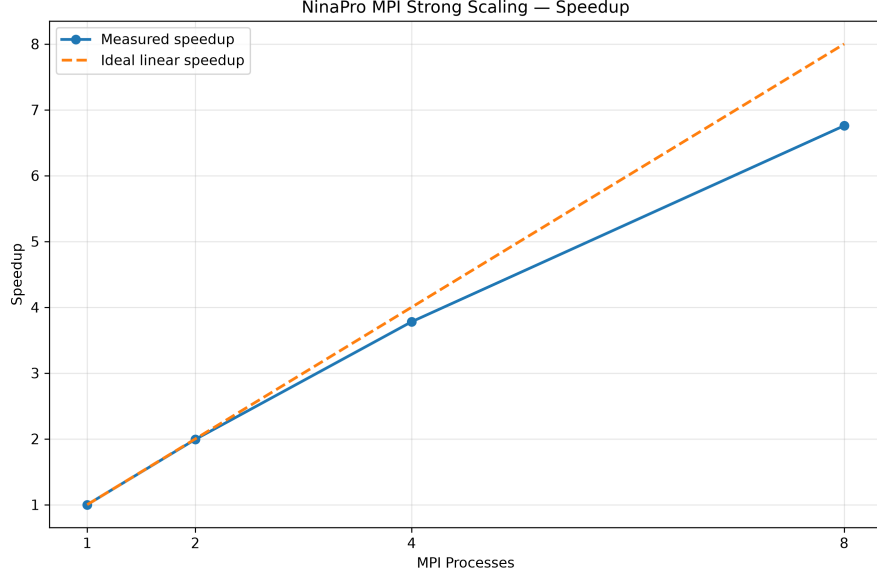


Fig. 3 Final NinaPro strong-scaling speedup on Bridges-2 relative to the single-process baseline. The dashed line represents ideal linear speedup.

4.6.2 Weak Scaling

The weak-scaling experiment increased the number of workload samples with MPI process count while maintaining approximately 5200 upper-triangular kernel-pair evaluations per rank.

Table 5 Final NinaPro kernel-pair weak-scaling results on Bridges-2.

MPI Processes	Samples	Kernel Pairs	Pairs/Rank	Mean Runtime (s)	Efficiency (%)
1	102	5253	5253.00	0.167482	100.00
2	144	10440	5220.00	0.123327	134.95
4	204	20910	5227.50	0.094989	175.46
8	289	41905	5238.12	0.084434	197.80

The number of kernel-pair calculations per process remained approximately constant across the four configurations. Mean runtime nevertheless decreased from 0.167482 s with one process to 0.123327 s with two processes, 0.094989 s with four processes, and 0.084434 s with eight processes.

The resulting calculated weak-scaling efficiencies exceeded 100% for the multi-process configurations. This superlinear result reflects the structure of the complete timed workflow rather than an increase in computational work performed by each process. Although kernel-pair evaluations per rank remain approximately constant, the timed region also includes quantum-state encoding. Because workload sample

count grows approximately with $\sqrt{P}$, the number of samples encoded by each rank decreases as process count increases. The reported weak-scaling efficiency should therefore be interpreted specifically as kernel-pair weak scaling of the complete implemented workflow.

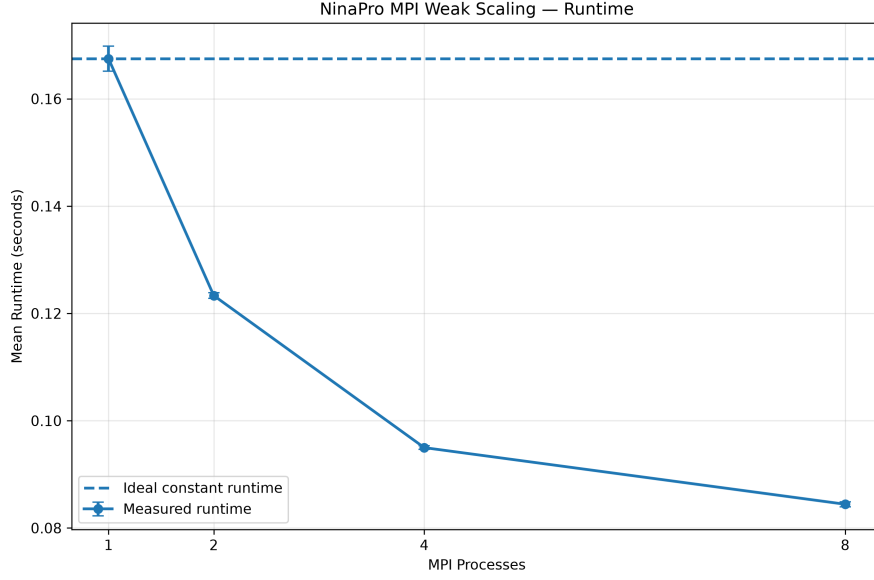


Fig. 4 Final NinaPro kernel-pair weak-scaling runtime on Bridges-2. The dashed line represents the single-process runtime expected under ideal constant-runtime weak scaling.

4.6.3 Multi-Node MPI Extension

The follow-on MPI experiment compared matched MPI process counts using single-node and multi-node placement. Table 6 summarizes the resulting runtimes.

Table 6 Matched single-node and multi-node MPI results for the fixed 102-sample NinaPro quantum-kernel workload on Bridges-2.

MPI Ranks	Nodes	Ranks/Node	Mean Runtime (s)	Std. (s)
8	1	8	0.025199	0.001515
8	2	4	0.026109	0.000258
16	1	16	0.014399	0.000372
16	2	8	0.020099	0.001448
32	1	32	0.009882	0.000359
32	4	8	0.014152	0.000246

At eight MPI processes, distributing the ranks across two nodes produced a mean runtime of 0.026109 s compared with 0.025199 s when all eight ranks were placed on one node. This represents a modest runtime increase of approximately 3.61%.

The effect of inter-node placement became substantially larger at the higher process counts. At sixteen MPI processes, the single-node configuration completed in 0.014399 s, whereas the two-node configuration required 0.020099 s, an increase of approximately 39.59%. At thirty-two processes, the single-node configuration completed in 0.009882 s compared with 0.014152 s when the same number of processes was distributed across four nodes, corresponding to approximately 43.21% higher runtime.

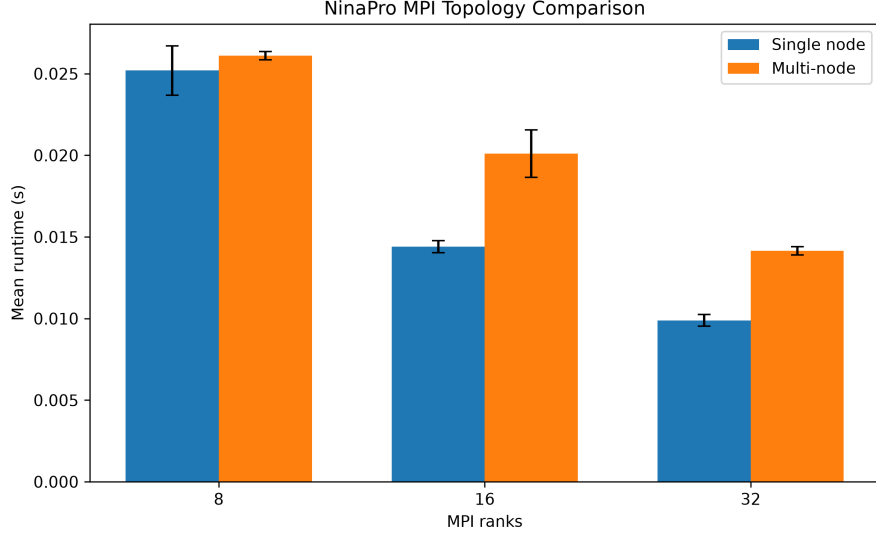


Fig. 5 Matched-rank MPI topology comparison for the fixed 102-sample NinaPro quantum-kernel workload on Bridges-2. Single-node configurations place all MPI ranks on one Regular Memory node. The multi-node configurations use two nodes for the 8- and 16-rank tests and four nodes for the 32-rank test. Error bars represent the standard deviation across five timed iterations.

Despite the inter-node overhead, increasing the total MPI process count continued to reduce runtime. Within the single-node configurations, increasing from 8 to 16 processes produced approximately a $1.75\times$ speedup, while increasing from 16 to 32 processes produced an additional $1.46\times$ speedup. Relative to the eight-process single-node configuration, thirty-two processes on one node reduced runtime by approximately $2.55\times$.

The multi-node configurations also benefited from increasing process count, but less efficiently. Runtime decreased from 0.026109 s at eight processes across two nodes to 0.020099 s at sixteen processes across two nodes and 0.014152 s at thirty-two processes across four nodes. These measurements show that additional MPI parallelism continued to improve performance, while inter-node communication reduced the benefit obtained from the additional processes.

4.6.4 Large Real-Data Multi-Node MPI Scaling

To evaluate multi-node MPI behavior under a substantially larger real-data workload, the NinaPro dataset was expanded from one to seven distinct non-overlapping 512-sample EMG windows per gesture/repetition segment. With 17 gestures and 6 repetitions, this produced 714 distinct real EMG windows and 255,255 upper-triangular quantum-kernel pairs.

Table 7 summarizes the measured results. Unlike the smaller 102-sample workload, matched single-node and multi-node configurations produced nearly identical runtimes.

Table 7 Large real-data MPI scaling results using 714 distinct NinaPro EMG windows and 255,255 quantum-kernel pairs on Bridges-2.

MPI Ranks	Nodes	Ranks/Node	Mean Runtime (s)	Speedup	Efficiency (%)
1	1	1	1.623986	1.000	100.00
8	1	8	0.281209	5.775	72.19
8	2	4	0.282289	5.753	71.91
16	1	16	0.187434	8.664	54.15
16	2	8	0.187977	8.639	54.00
32	1	32	0.136789	11.872	37.10
32	4	8	0.136360	11.910	37.22

At eight MPI processes, distributing the workload across two nodes changed mean runtime from 0.281209 s to 0.282289 s, a difference of approximately 0.38%. At sixteen processes, runtime changed from 0.187434 s on one node to 0.187977 s across two nodes, a difference of approximately 0.29%. At thirty-two processes, the four-node configuration completed in 0.136360 s compared with 0.136789 s on one node. This difference was only approximately 0.31% and was smaller than the observed run-to-run variation.

These results indicate that the inter-node communication overhead observed for the smaller workload was largely amortized when the number of real kernel-pair evaluations increased. The larger workload therefore scaled similarly whether a fixed number of MPI ranks was placed on one node or distributed across multiple Bridges-2 nodes.

4.6.5 Full-Capacity Real-Data MPI Scaling

A final MPI experiment used every complete non-overlapping 512-sample window available from the selected NinaPro recording. Across the 102 gesture-repetition segments, this produced 1,404 distinct real EMG windows and 986,310 upper-triangular quantum-kernel pair evaluations.

The first part of the experiment evaluated process-count scaling on a single Bridges-2 Regular Memory node using 1, 8, 16, 32, 64, and 128 MPI ranks. The second part held the total process count fixed at 128 ranks while distributing those processes across 1, 2, 4, 8, and 16 physical nodes.

Table 8 Full-capacity NinaPro MPI scaling using 1,404 distinct real EMG windows and 986,310 quantum-kernel pairs on Bridges-2.

MPI Ranks	Nodes	Ranks/Node	Mean Runtime (s)	Speedup	Efficiency (%)
1	1	1	4.198034	1.000	100.00
8	1	8	0.818761	5.127	64.09
16	1	16	0.565091	7.429	46.43
32	1	32	0.448628	9.357	29.24
64	1	64	0.437852	9.588	14.98
128	1	128	0.378437	11.093	8.67
128	2	64	0.389714	10.772	8.42
128	4	32	0.369076	11.374	8.89
128	8	16	0.367011	11.438	8.94
128	16	8	0.365481	11.486	8.97

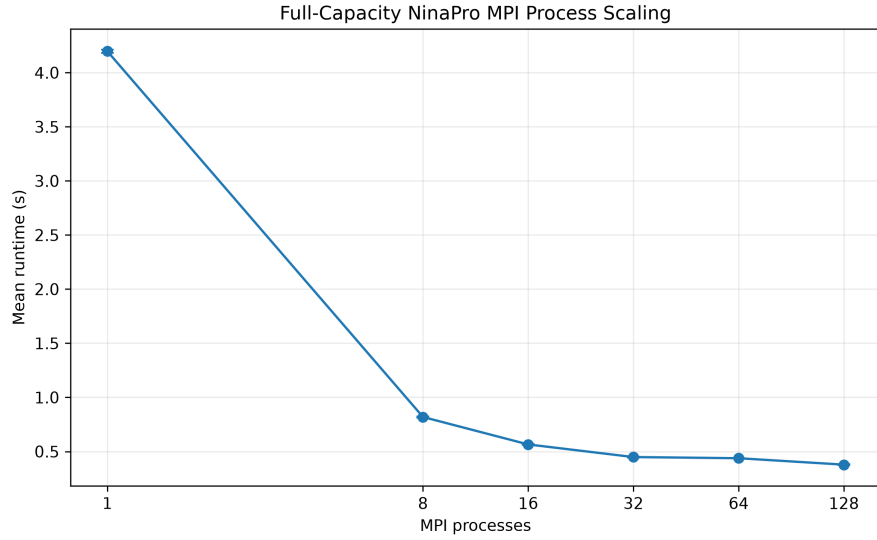


Fig. 6 Single-node process scaling for the full-capacity NinaPro workload containing 1,404 distinct real EMG windows and 986,310 quantum-kernel pairs. Error bars represent the standard deviation across five timed runs.

For the single-node process-scaling experiment, mean runtime decreased from 4.198034 s with one MPI process to 0.818761 s with eight processes, 0.565091 s with sixteen processes, 0.448628 s with thirty-two processes, 0.437852 s with sixty-four processes, and 0.378437 s with 128 processes. Relative to the single-process baseline, the 128-process single-node configuration achieved an $11.093\times$ speedup.

Parallel efficiency decreased as process count increased, from 64.09% at eight processes to 46.43% at sixteen, 29.24% at thirty-two, 14.98% at sixty-four, and 8.67% at 128 processes. The runtime reduction therefore continued through 128 processes, but with substantial diminishing returns.

Holding the total process count fixed at 128 provided a direct comparison of physical-node placement. Mean runtime was 0.378437 s on one node, 0.389714 s on two nodes, 0.369076 s on four nodes, 0.367011 s on eight nodes, and 0.365481 s on sixteen nodes.

The fastest measured configuration used 128 MPI processes distributed across 16 physical nodes, with eight ranks assigned to each node. This configuration achieved an $11.486\times$ speedup relative to the one-process baseline. Compared with the 128-process single-node configuration, the sixteen-node configuration reduced mean runtime by approximately 3.42%.

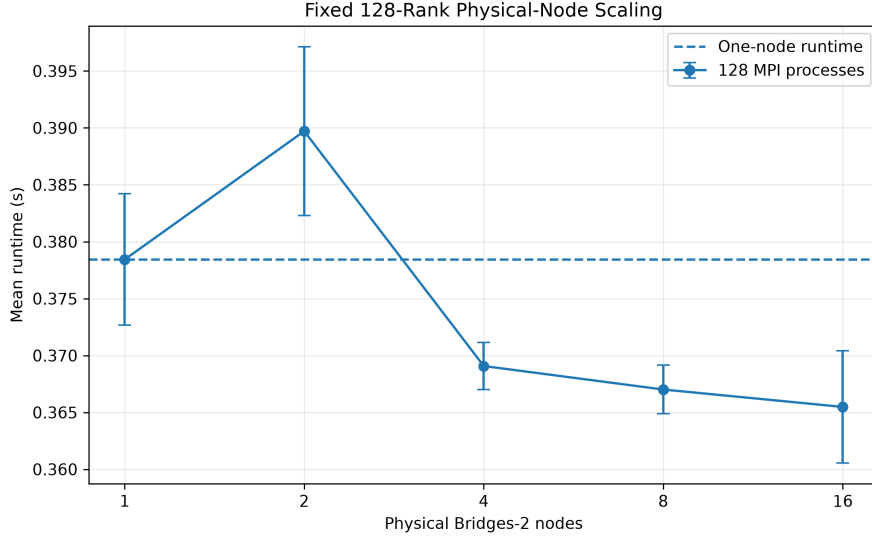


Fig. 7 Physical-node scaling with the total MPI process count fixed at 128 ranks. The same 1,404-window, 986,310-pair workload was distributed across 1, 2, 4, 8, and 16 Bridges-2 Regular Memory nodes. The dashed line represents the measured 128-rank one-node runtime.

The two-node configuration was approximately 2.98% slower than the one-node 128-rank configuration, while the four-, eight-, and sixteen-node configurations were approximately 2.47%, 3.02%, and 3.42% faster, respectively. These differences are modest compared with the overall process-count speedup and indicate that the larger workload was sufficient to largely amortize inter-node communication overhead.

4.6.6 Bridges-2 V100 Accelerator Benchmark

The final accelerator experiment was performed on an exclusive Bridges-2 GPU node equipped with NVIDIA Tesla V100-32GB GPUs. Qiskit Aer statevector simulation was evaluated on both the CPU and a single V100 GPU. The Aer GPU tensor-network method was also evaluated using the installed cuQuantum software stack. Each configuration used one untimed warm-up followed by five timed executions, and all resulting quantum-kernel matrices passed numerical validation.

For the complete 102-sample, eight-qubit NinaPro kernel, the CPU statevector backend required a mean runtime of 7.132028 s. The V100 statevector backend required 7.210690 s, corresponding to a CPU-relative speedup of $0.989\times$. The GPU tensor-network backend required 8.277748 s, corresponding to a CPU-relative speedup of $0.862\times$. Therefore, GPU acceleration did not improve performance for the complete eight-qubit kernel workload.

Table 9 Final Bridges-2 backend comparison for the 102-sample, eight-qubit NinaPro quantum kernel.

Backend	Mean Runtime (s)	Std. (s)	Speedup vs. CPU
CPU statevector	7.132028	0.044020	1.000000
V100 statevector	7.210690	0.034396	0.989091
V100 tensor network	8.277748	0.026685	0.861590

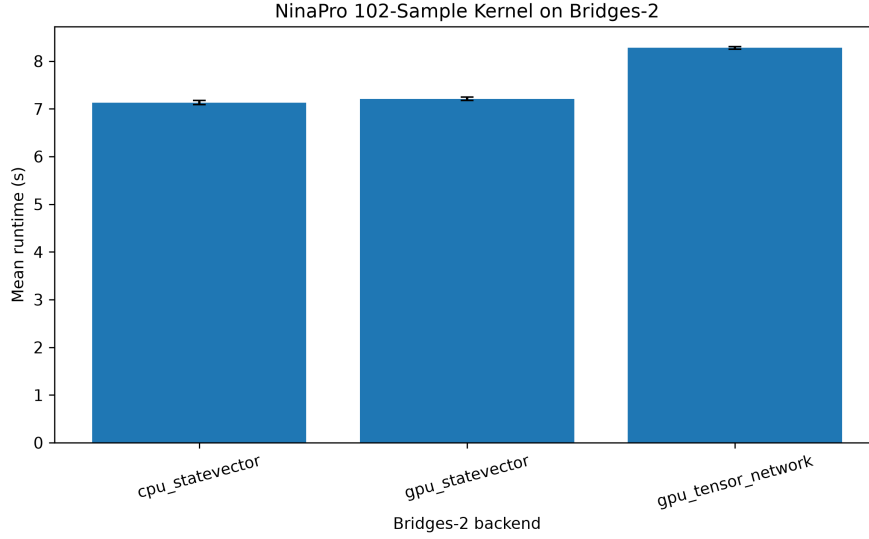


Fig. 8 Mean runtime of the final 102-sample NinaPro quantum-kernel workload using Aer CPU statevector, V100 GPU statevector, and V100 GPU tensor-network simulation on Bridges-2. Error bars represent the standard deviation across five timed runs.

A separate controlled scaling experiment evaluated 4, 6, 8, 10, 12, 14, and 16 qubits using eight deterministic feature vectors per circuit size. CPU and V100 statevector runtimes remained similar from 4 through 14 qubits, with the CPU retaining a small performance advantage at each of those tested sizes. At 16 qubits, however, mean CPU runtime increased to 0.634693 s while V100 statevector runtime was 0.594701 s, producing a measured $1.067\times$ GPU speedup.

The GPU tensor-network method remained slower than CPU statevector simulation throughout the tested range. Its CPU-relative speedup decreased from $0.883\times$ at four qubits to $0.769\times$ at sixteen qubits.

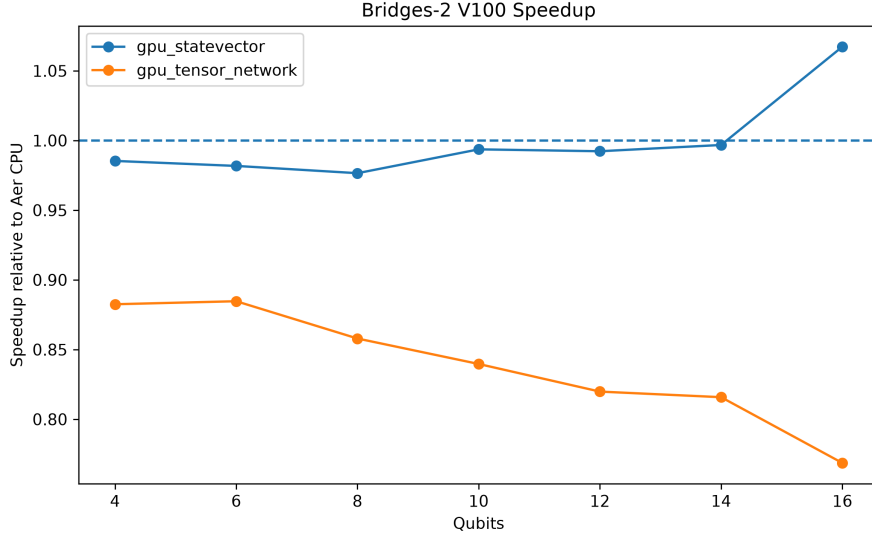


Fig. 9 Bridges-2 V100 accelerator speedup relative to Aer CPU statevector simulation from 4 to 16 qubits. The dashed line at $1.0\times$ represents equal CPU and GPU performance. V100 statevector simulation exceeds CPU performance only at the largest tested circuit size of 16 qubits.

These measurements indicate that accelerator overhead dominates the smaller quantum-simulation workloads tested here. As the simulated statevector grows, the relative cost of GPU overhead becomes less significant, and a performance advantage begins to appear at the largest tested circuit size. Because the experiment evaluates only discrete circuit sizes through 16 qubits, the results identify a measured advantage at 16 qubits rather than establishing a general CPU/GPU crossover threshold.

5 Discussion

5.1 CPU and GPU Simulation Behavior

The local CPU/GPU experiments demonstrate that accelerator use does not automatically improve quantum-simulation performance. For the complete eight-qubit NinaPro quantum-kernel workload, the CPU implementation completed the benchmark faster than the local RTX 2080 SUPER GPU implementation. In contrast, the controlled qubit-scaling experiment showed a change in relative performance at 16 qubits, where the GPU completed the tested workload approximately $2.44\times$ faster than the CPU.

These results suggest that the usefulness of GPU acceleration depends strongly on workload size. At lower qubit counts, simulator setup, kernel launch, memory transfer, and other accelerator overheads may represent a substantial fraction of total runtime.

As the size of the simulated state space increases, the amount of parallel numerical work also increases, allowing GPU execution to become more competitive.

The measured crossover should be interpreted as specific to the tested hardware, circuit construction, simulator implementation, and benchmark configuration. In particular, the CPU and GPU paths use different simulation software, and the local GPU measurements were obtained using an NVIDIA GeForce RTX 2080 SUPER. The results therefore characterize the behavior of the implemented workflow rather than establishing a universal qubit threshold at which GPU simulation becomes preferable.

The comparatively large runtime variation observed for the 16-qubit CPU measurement also warrants caution when interpreting the exact magnitude of the measured speedup. The important result is the observed change in relative CPU/GPU behavior as circuit size increased, rather than the identification of a fixed crossover point.

The Bridges-2 accelerator experiment reinforces the workload-size dependence observed during local GPU testing. For the complete eight-qubit NinaPro kernel, V100 statevector simulation did not outperform CPU statevector simulation, with mean runtimes of 7.210690 s and 7.132028 s, respectively. This indicates that the computational work at eight qubits remained too small to offset GPU execution overhead.

In the controlled qubit-scaling experiment, CPU and V100 statevector performance remained close from 4 through 14 qubits. At the largest tested size of 16 qubits, the V100 statevector backend became faster, producing a measured $1.067\times$ speedup relative to the Aer CPU backend. This result is consistent with the expectation that accelerator hardware becomes more useful as statevector size and simulation work increase.

The GPU tensor-network method did not provide a performance advantage for the tested circuits. Tensor-network performance depends strongly on circuit structure, entanglement, contraction strategy, and simulation size; therefore, the observed result should not be interpreted as a general comparison between tensor-network and statevector simulation. Rather, the shallow manual ZZFeatureMap circuits and modest 4–16-qubit range tested here did not provide a workload for which the additional tensor-network machinery produced a runtime benefit.

The Bridges-2 results should also be interpreted separately from the local RTX 2080 SUPER experiment. The two environments use different hardware and simulation software stacks, so absolute runtime values are not treated as a direct GPU-to-GPU hardware comparison. Instead, both experiments are used to evaluate how accelerator usefulness changes as the quantum simulation workload increases.

5.2 Numerical Consistency Across Simulation Backends

The CPU and GPU implementations produced closely matching quantum-kernel matrices, with a maximum absolute element-wise difference of approximately 4.77×10^{-7} and a mean absolute difference of approximately 5.04×10^{-9} . Both implementations also satisfied the expected kernel properties, including symmetry, a unit diagonal, finite values, and entries within the interval $[0, 1]$.

This numerical agreement is important because the GPU implementation is not simply a different execution mode of the same simulator. The manually constructed

Qiskit feature-map circuit is translated into a separate qsim-based GPU representation. Agreement between the two resulting kernel matrices therefore provides evidence that the translation preserves the intended quantum-kernel computation within normal floating-point numerical variation.

The validation also supports the use of the same biomedical feature vectors across the subsequent performance experiments. Performance comparisons are therefore based on implementations that have first been checked for numerical consistency rather than on runtime alone.

5.3 Effect of Depolarizing Noise

The controlled depolarizing-noise experiment produced a monotonic decrease in mean state fidelity as the noise probability increased from $p = 0.001$ to $p = 0.050$. Mean fidelity decreased from approximately 0.99996 to approximately 0.99804 over the tested range, while the variation in fidelity across the 102 samples increased with noise strength.

Although the numerical decrease in mean fidelity is relatively small, the trend is consistent across the tested noise levels. These results demonstrate that the implemented workflow can quantify degradation of the encoded NinaPro states under a controlled quantum-noise model. They should not, however, be interpreted as a characterization of a particular physical quantum processor. The experiment uses a simplified depolarizing model applied within classical simulation.

Mean runtime remained close to 0.042 seconds per sample across all five noise strengths. Within the tested range, increasing the depolarizing probability therefore changed the resulting quantum states without producing a comparable increase in simulation time. This result is specific to the eight-qubit density-matrix workload and the tested Aer configuration.

The quantum-noise experiment is also distinct from the differential-privacy experiments conducted during earlier stages of the broader DREU-QIS project. The parameter p represents simulated quantum gate noise, whereas the differential-privacy parameter ϵ controls a privacy mechanism. The two parameters describe different experimental phenomena and are not directly comparable.

5.4 MPI Scaling and HPC Execution

The Bridges-2 experiments demonstrate that the pairwise quantum-kernel workload can be parallelized effectively using workload-level MPI. In the original 102-sample strong-scaling experiment, mean runtime decreased from 0.168016 s with one process to 0.024862 s with eight processes, producing a measured $6.76\times$ speedup. Parallel efficiency remained 99.62% at two processes, 94.48% at four processes, and 84.47% at eight processes.

The accompanying weak-scaling experiment maintained approximately 5200 kernel-pair evaluations per rank. Calculated efficiencies exceeded 100% because the complete timed workflow includes both pair evaluation and quantum-state encoding, and the amount of state-encoding work assigned to each rank decreases as process

count increases. These values should therefore be interpreted as kernel-pair weak scaling of the implemented workflow rather than conventional superlinear scaling of a strictly constant workload per process.

The first multi-node topology experiment used the small 102-sample workload. At matched process counts, distributing ranks across multiple physical nodes increased runtime, particularly at sixteen and thirty-two processes. This indicated that inter-node communication represented a significant fraction of the total execution time when only 5253 kernel-pair evaluations were available.

Increasing the workload to 714 distinct real EMG windows and 255,255 kernel-pair evaluations substantially changed this behavior. Matched single-node and multi-node configurations differed by less than 0.4% at 8, 16, and 32 MPI processes, indicating that the increased computation was sufficient to largely amortize the inter-node communication cost.

The final full-capacity experiment extended this trend using 1,404 distinct non-overlapping real EMG windows and 986,310 kernel pairs. On a single node, runtime decreased from 4.198034 s with one MPI process to 0.378437 s with 128 processes, corresponding to an $11.093\times$ speedup. Parallel efficiency, however, declined to 8.67% at 128 processes, demonstrating substantial diminishing returns at high process counts.

The fixed-128-rank topology experiment provides the clearest isolation of physical-node effects. Moving from one to two nodes increased runtime by approximately 2.98%, while distributing the same 128 ranks across four, eight, and sixteen nodes reduced mean runtime by approximately 2.47%, 3.02%, and 3.42%, respectively, relative to the one-node configuration. The fastest measured topology used sixteen nodes and eight ranks per node, with a mean runtime of 0.365481 s.

The modest size of these topology differences compared with the much larger process-count speedup is important. The results do not show that adding physical nodes alone produces large acceleration. Rather, they show that once the workload becomes sufficiently large, the cost of inter-node communication can be amortized to the point that distributing a fixed number of ranks across several physical nodes is competitive with, and in some tested configurations slightly faster than, concentrating all ranks on one node.

Across the 102-, 714-, and 1,404-window experiments, the overall pattern is therefore consistent: workload size strongly affects the usefulness of distributed-memory execution. Small workloads expose communication overhead more clearly, while larger workloads provide enough independent pairwise work to make multi-node execution increasingly practical. At the same time, the declining parallel efficiency observed at high MPI process counts shows that communication, synchronization, state exchange, result gathering, and other non-parallel costs ultimately limit strong-scaling performance.

5.5 Biomedical Workload and Scope

The use of NinaPro DB2 provides a real biomedical signal-processing workload for evaluating the computational framework. Rather than benchmarking only randomly generated numerical vectors or synthetic circuit inputs, the final kernel is constructed from features derived from recorded surface-EMG signals.

The present study does not attempt to evaluate gesture-classification accuracy or clinical performance. Gesture labels are used to identify the corresponding recording segments and construct a reproducible set of samples, but no classifier-performance claim is made from the resulting quantum kernel.

Similarly, the use of eight PCA-derived features is primarily a computational design choice that allows the processed biomedical data to be mapped directly to an eight-qubit feature map. The experiment therefore evaluates the computational behavior of a quantum-kernel workflow on biomedical data rather than demonstrating that eight quantum features represent an optimal classification model.

5.6 Limitations

Several limitations should be considered when interpreting the results.

First, the biomedical benchmark uses a single NinaPro DB2 subject and a single exercise/acquisition file. The primary workload contains one selected window from each of 17 gesture classes and six repetitions. The balanced larger workload contains seven distinct non-overlapping windows from each of those gesture-repetition segments, while the final full-capacity workload contains every complete non-overlapping 512-sample window available from those same segments, for a total of 1,404 windows. These larger workloads increase the amount of real recorded EMG data used for computation but do not represent additional subjects or independent gesture repetitions, and they should not be interpreted as representative of the complete NinaPro DB2 population.

Second, larger weak-scaling workloads reuse the same 102 processed feature vectors deterministically. The 144-, 204-, and 289-sample workloads therefore represent increased computational workloads rather than additional independent biomedical observations.

Third, all quantum results in this study are produced through classical quantum simulation. No claim of quantum computational advantage is made, and the measured CPU, GPU, and MPI behavior reflects the cost of simulating the quantum workload on classical hardware.

Fourth, the local CPU and GPU implementations use different simulation software. Although numerical agreement was explicitly validated, performance differences reflect both the underlying hardware and the implementation details of the respective simulators.

Fifth, the depolarizing-noise experiment uses a controlled simulation model and does not reproduce the complete error characteristics of a specific physical quantum processor.

Finally, the weak-scaling experiment maintains an approximately constant number of kernel-pair calculations per MPI rank rather than a strictly constant amount of total timed work per rank. The timed workflow also includes quantum-state encoding, state exchange, and result gathering. The resulting weak-scaling efficiencies above 100% should therefore be interpreted in the context of this workload design rather than as evidence of universally superlinear MPI scaling.

The MPI experiments remain limited in dataset scope despite the increased computational workload. The primary kernel contains 102 samples and 5253 upper-triangular pair evaluations, the balanced larger workload contains 714 distinct windows and 255,255 pairs, and the final full-capacity workload contains 1,404 distinct non-overlapping windows and 986,310 pairs. All three workloads are derived from the same Subject 1, Exercise 1, Acquisition 1 recording.

Multi-node execution was evaluated through 16 Bridges-2 Regular Memory nodes and 128 MPI processes. The observed topology behavior should therefore not be generalized directly to the complete NinaPro DB2 dataset, substantially higher-qubit simulations, larger node counts, or different communication patterns. The MPI implementation parallelizes independent quantum-state encoding and quantum-kernel tasks; it does not distribute a single quantum statevector across multiple nodes.

6 Conclusion

This work developed and evaluated a reproducible high-performance computing workflow for simulated quantum-kernel calculations using biomedical EMG data. The final experimental pipeline connects NinaPro DB2 preprocessing, classical feature extraction, dimensionality reduction, manual quantum feature-map construction, quantum-kernel evaluation, CPU and GPU simulation, controlled noise characterization, and MPI-based workload distribution.

The NinaPro benchmark produced 102 eight-dimensional quantum feature vectors derived from 12-channel surface-EMG recordings. These vectors were used to construct a complete 102×102 quantum-kernel workload based on an eight-qubit manually constructed feature map. Independent CPU and CUDA-capable GPU implementations produced numerically consistent kernel matrices, with a maximum absolute difference of approximately 4.77×10^{-7} .

Local performance measurements showed that the CPU completed the full eight-qubit NinaPro kernel benchmark faster than the GPU. However, controlled qubit-scaling experiments demonstrated that the relative performance changed as circuit size increased. At 16 qubits, the local GPU completed the tested simulation approximately $2.44\times$ faster than the CPU. These measurements show that accelerator benefit depends on workload size and that small quantum simulations do not necessarily provide enough computational work to offset GPU overhead.

The controlled depolarizing-noise experiment further demonstrated that the framework can evaluate changes in simulated quantum-state fidelity. Mean fidelity decreased monotonically across the tested noise strengths, from approximately 0.99996 at $p = 0.001$ to approximately 0.99804 at $p = 0.050$, while average simulation runtime remained close to 0.042 seconds per sample.

The project also demonstrated distributed-memory execution on the Bridges-2 high-performance computing system. For the fixed 102-sample NinaPro workload, single-node strong scaling reduced mean runtime from 0.168016 s with one MPI process to 0.024862 s with eight processes, corresponding to a $6.758\times$ speedup and 84.47% parallel efficiency. Weak-scaling experiments further demonstrated that the independent pairwise structure of the quantum kernel can be distributed across MPI

ranks, although calculated efficiencies above 100% reflect the mixed state-encoding and kernel-pair work contained in the timed workflow.

A first multi-node experiment using the 102-sample workload showed measurable communication overhead when matched process counts were spread across additional nodes. Increasing the workload to 714 distinct real EMG windows and 255,255 kernel pairs largely removed this penalty, with matched single-node and multi-node runtimes differing by less than 0.4%.

The final full-capacity MPI experiment used 1,404 distinct non-overlapping real EMG windows and 986,310 kernel pairs. On one node, increasing process count from one to 128 reduced mean runtime from 4.198034 s to 0.378437 s, corresponding to an $11.093\times$ speedup. When the same 128 MPI ranks were distributed across 16 Bridges-2 nodes, mean runtime decreased further to 0.365481 s, producing the fastest measured configuration and an overall $11.486\times$ speedup relative to the single-process baseline.

These results show that workload-level MPI can effectively exploit the independent pairwise structure of quantum-kernel construction, while also demonstrating two important scaling limits. First, increasing MPI process count produces diminishing returns as communication and synchronization overhead become more significant. Second, the effect of distributing ranks across physical nodes depends strongly on workload size: multi-node overhead was clearly visible for the smallest workload, approximately neutral for the 714-window workload, and modestly favorable in several configurations of the 1,404-window workload.

The final Bridges-2 accelerator experiment additionally evaluated Aer CPU statevector, V100 GPU statevector, and V100 GPU tensor-network simulation. For the complete eight-qubit NinaPro kernel, accelerator overhead prevented either GPU configuration from outperforming CPU execution. In the controlled qubit-scaling experiment, however, V100 statevector simulation became faster than CPU statevector simulation at the largest tested size of 16 qubits, achieving a measured $1.067\times$ speedup. These results further demonstrate that accelerator effectiveness depends strongly on simulation scale and workload characteristics.

Overall, the work demonstrates a complete experimental path from real biomedical signals to a simulated quantum-kernel workload that can be validated and benchmarked across CPU, GPU, noisy-simulation, and distributed HPC configurations. The results do not establish quantum advantage or biomedical classification superiority; rather, they characterize the computational requirements and scaling behavior that must be understood when developing larger quantum-machine-learning simulation studies.

Author Contributions

Kyle Hartness developed the HPC benchmarking and parallelization workflow, implemented the NinaPro biomedical workload, developed and validated the local CPU and GPU quantum-kernel simulation paths, performed the qubit-scaling and depolarizing-noise experiments, developed the MPI strong-, weak-, and multi-node scaling benchmarks, designed and executed the 714-window and 1,404-window real-data MPI extensions through 128 MPI ranks and 16 physical Bridges-2 nodes, configured and

evaluated the Bridges-2 V100 accelerator environment, and analyzed the resulting performance data.

Edmund Bombardieri contributed to the quantum-computing component of the broader DREU-QIS project, including development and evaluation of the quantum feature-map and quantum-kernel methodology that informed the workload used in this study.

Sofia Furda contributed to the broader project architecture, including the privacy-aware and distributed-processing components developed in parallel with the HPC benchmarking work.

All authors reviewed the project direction and contributed to the broader DREU-QIS research effort.

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Data Availability

The biomedical workload used in this study was derived from the publicly available NinaPro DB2 surface-electromyography dataset. Raw NinaPro data are not redistributed with the QuantumHPC repository. The preprocessing and benchmark code required to construct the experimental workload from an appropriately obtained NinaPro DB2 file is included in the project repository.

Code Availability

The software developed for this project is maintained in the QuantumHPC project repository. The repository includes the NinaPro preprocessing pipeline, manual quantum feature-map implementation, quantum-kernel evaluation code, CPU and GPU simulation interfaces, depolarizing-noise benchmark, MPI strong- and weak-scaling benchmarks, SLURM execution scripts, and analysis utilities used to generate the reported results.

The repository will be referenced by its final public archival URL or release identifier in the completed manuscript.

Competing Interests

The authors declare no competing interests.

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